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Feature Articles

SHIFTS IN SEASONAL CLIMATIC SUITABILITY FOR CASSIN'S SPARROW (*PEUCAEA CASSINII*) OVER FOUR DECADES

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ABSTRACT–Cassin's Sparrow (*Peucaea cassinii*) is a resident species of the southwestern United States and northern Mexico. Like many aridland birds, Cassin's Sparrow is thought to be an opportunistic breeder, its movements and other aspects of its reproductive cycle influenced by favorable conditions rather than a fixed schedule. In this study, we used retrospective ecological niche modeling, environmental data from NASA's MERRA-2 reanalysis, and 40 years of eBird observations to examine short-term and long-term trends in climatic suitability for the species during its breeding season. We found a consistent east-to-west progression of climatic suitability for Cassin's Sparrow during the summer months, with an annual average northwesterly movement of ~613 km over the 40-year study period. These shifts, driven largely by surface wind speed and variability in temperature and precipitation, have diminished over time from ~922 km over the summer season in the mid-1980s to ~304 km in recent years. Spatial analysis revealed that the amount of change in suitability over the course of the breeding season is increasing in central and western parts of Cassin's Sparrow's range and decreasing in southeastern Texas. This regional variation aligns with reported changes in breeding phenology, suggesting that the magnitude of seasonal change in suitability itself might have an important influence on the species' breeding cycle. Our findings further suggest that Cassin's Sparrow's response to long-term climate change might increasingly depend on adapting to regional conditions rather than the opportunistic movements once thought to define its breeding strategy.

RESUMEN–El Gorrión de Cassin (*Peucaea cassinii*) es una especie residente del suroeste de los Estados Unidos y del norte de México. Como muchas aves de tierras áridas, se considera un reproductor oportunista, cuyos desplazamientos y otros aspectos de su ciclo reproductivo están influenciados por condiciones favorables en lugar de por un calendario fijo. En este estudio, utilizamos modelos retrospectivos de nicho ecológico, datos ambientales del reanálisis MERRA-2 de la NASA y 40 años de observaciones de eBird para examinar las tendencias a corto y largo plazo en la idoneidad climática para la especie durante su temporada reproductiva. Detectamos una progresión constante de la idoneidad climática para el Gorrión de Cassin de este a oeste durante los meses de verano, con un desplazamiento promedio anual hacia el noroeste de ~613 km a lo largo del periodo de estudio de 40 años. Estos desplazamientos, impulsados principalmente por la velocidad del viento en la superficie y la variabilidad de la temperatura y la precipitación, han disminuido con el tiempo, pasando de ~922 km en la temporada de verano a mediados de la década de 1980 a ~304 km en años recientes. El análisis espacial reveló que la magnitud del cambio en la idoneidad a lo largo de la temporada reproductiva está aumentando en las zonas central y occidental del área de distribución de la especie, pero está disminuyendo en el sureste de Texas. Esta variación regional coincide con cambios documentados en la fenología reproductiva, lo que sugiere que la magnitud del cambio estacional en la idoneidad podría ejercer una influencia importante en el ciclo reproductivo de la especie. Nuestros hallazgos sugieren, además, que la respuesta del Gorrión de Cassin al cambio climático a largo plazo podría depender cada vez más de la adaptación a las condiciones regionales, en lugar de los desplazamientos oportunistas que antes se consideraban característicos de su estrategia reproductiva.

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Cassin's Sparrow (*Peucaea cassinii* Woodhouse, 1852) is a native species of the arid shrub-grasslands of the southwestern United States (U.S.), southern Great Plains, and northern Mexico (Dunning et al., 2020). The species is migratory; constructs nests near the ground; and is notable for its secretive nature, highly developed flight song, and desert adaptations, especially its ability to move and breed opportunistically (Hubbard, 1977; A. R. Phillips, 1944; Ruth, 2000). In opportunistic breeders, a bird's geographic range, annual migration, seasonal movements, reproductive phases, clutch sizes, number of broods, and other aspects of the breeding ecology of a bird are based on favorable environmental conditions, rather than a fixed schedule (Duursma et al., 2017; Newton, 2024; J. B. Williams & Tieleman, 2005). This plasticity is adaptive given the highly variable and unpredictable nature of arid ecosystems and the physiological stresses placed on birds in these environments (Ma et al., 2023). Because of its apparent sensitivity and responsiveness to environmental conditions, Cassin's Sparrow is often used as a representative species in studies of aridland ecosystems (Forrest, 2022; Gordon, 2000; Grand et al., 2019; Roberts et al., 2012, 2017; Salas et al., 2017; G. M. Visser, 2021; Wilsey et al., 2019). Cassin's Sparrow also has been used as an indicator species for the long-term monitoring of grassland biodiversity and ecosystem health (Lynn, 2006; Ruth, 2000; U.S. Fish and Wildlife Service, 2012, 2016).

Changing climatic conditions are known to affect opportunistic behaviors in birds (Neate-Clegg et al., 2024; Skagen & Adams, 2012). Recent work has revealed a 40-year shift in year-round climatic suitability for Cassin's Sparrow (Schnase et al., 2024); however, details about the climatic influences driving Cassin's Sparrow's seasonal opportunistic behaviors are poorly understood. Studies have shown that the breeding ecology of the species is linked to widespread, summertime weather patterns and localized, ground-level climatic conditions (Dunning et al., 2020; Ruth, 2000; Schnase et al., 1991, 2024; F. C. Williams & LeSassier, 1968). The breeding season begins in southeastern Texas in early spring and ends in southeastern Arizona in late summer (Ohmart, 1966, 1969; A. R. Phillips, 1944; Ruth, 2000; Schnase et al., 2025). Regional precipitation appears to be an important driver, presumably due to its influence on vegetation availability and insect abundance (Dunning et al., 2020; Ohmart, 1969; Schnase et al., 2024; F. C. Williams & LeSassier, 1968). In fact, field studies over the past century suggest that Cassin's Sparrow might be an itinerant breeder, so responsive to regional rainfall patterns that birds make seasonal, inter-clutch moves within their range to find optimal conditions for breeding (Dunning et al., 2020; Ohmart, 1969; Ruth, 2000; F. C. Williams & LeSassier, 1968).

Understanding how long-term climate change affects behaviors in opportunistically breeding birds is particularly challenging, because the influences of a changing climate play out across two interrelated spatiotemporal frames: (1) seasonal changes, largely driven by weather and local environmental conditions, that can alter resource availability and influence the summertime movements and breeding activity of a species and (2) decadal changes, driven by global climate change, that can shift seasonal conditions altogether and alter long-term migration patterns and cycles of ecosystem productivity (Bateman, Pidgeon, Radeloff, Flather, et al., 2016; Fandos et al., 2020; Pascoe et al., 2021; Renner & Zohner, 2018; Robertson et al., 2024; M. E. Visser et al., 2012). While most studies in this area provide valuable information about the magnitude and impact of decadal change, they often lack the temporal resolution needed to evaluate the intra-annual changes to which a vagile, opportunistically breeding bird is responding. Importantly, these seasonal variations can have a stronger influence on the immediate environment and breeding cycle of aridland birds than do long-term average conditions (Bateman, Pidgeon, Radeloff, Flather, et al., 2016; Rushing et al., 2017; Schuster et al., 2019).

In this study, we draw on 40 years of citizen scientist observations and environmental data from NASA's Earth system models to improve our understanding of the climatic drivers influencing Cassin's Sparrow's breeding ecology. As described below, we use a multiscale adaptation of retrospective ecological niche modeling (rENM) in our analysis. rENM combines historical occurrence records and environmental data to estimate the relative suitability of conditions across space and time, producing time series that characterize the spatiotemporal dynamics of suitable conditions for a species (Elith & Leathwick, 2009; Schnase et al., 2024). These historical patterns can shed light on biological questions and the present and future conservation status of a species (Johnston et al., 2020). While similar studies often rely solely on temperature and precipitation to represent changing climatic conditions, we employ a combination of time-specific macroclimatic and microclimatic drivers in our models. In a further refinement, we use rENM to examine climatic suitability changes for Cassin's Sparrow at both seasonal and decadal scales.

This study addresses four primary questions: (1) What are the seasonal patterns of climatic suitability for Cassin's Sparrow? (2) How have these patterns changed over the past four decades? (3) What are the key climatic variables driving these seasonal dynamics? and (4) What insights into the breeding ecology of the species can be gained through this approach? By addressing these questions, we provide a more comprehensive understanding of both seasonal weather patterns and long-term climatic changes influencing Cassin's Sparrow. Beyond advancing our understanding of this single species, this work offers a promising approach to studying the multiscale dynamics of climatic influences on grassland and aridland birds more broadly.

MATERIALS AND METHODS

Approach to the study–Our study included both a seasonal analysis and a long-term, decadal analysis, each having two components. To analyze seasonal changes, we first created climatic suitability models for each 2-week interval during the breeding season months of April through August for each 5-year span from 1985 through 2024. We then used the Theil-Sen median slope estimate to determine the seasonal suitability trends across these biweekly models (Sen, 1968; Theil, 1950). These trend maps gave us a pixel-level view of the seasonal shifts in climatic suitability across the study area for each of the eight 5-year spans of the 40-year

study. Second, we calculated the weighted centroids for climatic suitability for each biweekly interval. Weighted centroids represent the geometric center of all pixels in the studied area, weighted by their suitability index. We chose weighted centroids for this purpose because of their ability to account for information derived from across the entire modeled range of a species (Huang et al., 2023; Vanderwal et al., 2013). These centroids allowed us to plot biweekly, study area-wide changes in the location of this overall estimate of suitability. We analyzed centroid movement by regressing the centroid coordinates against the 2-week intervals of each 5-year span. We also determined the traversal distance and bearing of these seasonal shifts, which allowed us to compute an overall direction and kilometers per week traversal speed for each 5-year span.

We used results from this seasonal analysis in our decadal analysis, which also consisted of two elements. First, we regressed the eight traversal distances from the seasonal analysis against 5-year spans to quantify 40-year changes in the seasonal shift distance. We then tested for long-term spatial trends in the magnitude of seasonal changes. We did this by building a set of eight layers representing the difference between the first and last biweekly climatic suitability model in each 5-year span. After that, we applied the Thiel-Sen median slope estimate to these layers, which gave us a pixel-level, range-wide view of the changing magnitude of seasonal shifts in suitability over the past 40 years. We looked at this trend to understand how the size of the seasonal suitability shifts might vary across regions as a result of long-term climatic change. We fitted linear regressions under a Bayesian framework using the R rstanarm package (Goodrich et al., 2020). We ran three Markov chains with 100,000 iterations each, including a burn-in of 10,000 iterations and a thinning interval of two, using weakly informative priors for all parameters (Gelman et al., 2017). We extracted mean estimates and regression coefficients (β) from the posterior distributions, used 95% credible intervals (CIs) to assess parameter uncertainty, used Bayesian R^2 to evaluate model fit and its associated uncertainty (Gelman et al., 2019), and report means ± SE for both sample-based and posterior distributions throughout.

Preparing the input data–For species occurrences, we obtained a total of 61,192 unique time-stamped Cassin's Sparrow point locations for the years 1985 to 2024 from the eBird database (Cornell Lab, 2025). We removed redundant and incomplete records, then binned the records into biweekly intervals across the breeding season months of April through August in each of the eight 5-year spans of the study. This binning resulted in 10 biweekly intervals per 5-year span, with days 1-15 of each month constituting a first bin, days 16-30 or 31 constituting a second bin.

Annual eBird records for Cassin's Sparrow have increased approximately 20-fold over the past two decades, a trend driven largely by increased observer participation rather than changes in the species' underlying population dynamics. This elevated sampling intensity can introduce temporal and spatial biases that distort species-environment relationships in ecological niche models (ENMs) (Boakes et al., 2010; Boria et al., 2014; Inman et al., 2021; S. J. Phillips et al., 2009). However, time-slicing and the temporal matching of occurrence records with environmental variables, as implemented in the rENM approach used here, have been shown to effectively reduce the influence of temporal bias (Ingenloff & Peterson, 2021; Nadeau et al., 2017; S. J. Phillips et al., 2009; Pinilla-Buitrago, 2025).

Similarly, spatial filtering is a common and effective approach for reducing spatial bias in species occurrence data (Baker et al., 2024; Boria et al., 2014; Fourcade et al., 2014; Inman et al., 2021; Kramer-Schadt et al., 2013). To address this issue, we used a two-step filtering process. First, to minimize the likelihood of double-counting individual birds, we used the R spThin package to thin occurrence records in each biweekly interval to non-overlapping observations within an 8 km (~5 mile) buffer (Aiello-Lammens et al., 2019). We chose an 8 km buffer because it substantially exceeds the documented home-range radius of Cassin's Sparrow (Cooper et al., 2014; Gordon, 2000; Maxwell, 1979; Ruth, 2000; Schnase, 1984). Second, to ensure uniform sample sizes across time, and to limit record densities that can degrade model performance, we randomly subsampled 250 records per biweekly interval from the thinned dataset (Pearson et al., 2007; S. J. Phillips et al., 2017). Across exploratory model runs, this two-step spatial filtering approach produced the most consistently accurate models over the 40-year time series (Schnase et al., 2024).

For environmental variables, we used 14 variables from NASA's Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2) reanalysis (Bosilovich et al., 2016; Gelaro et al., 2017) that are known to be important determinants of climatic suitability for Cassin's Sparrow (Schnase et al., 2024; Schnase & Carroll, 2022) (Tables 1, 2). With reanalysis, a data assimilation system is used to reprocess historical meteorological observations from satellite, airborne, and in-situ sensors. The system typically employs an iterative, time-stepping procedure in which an existing modeled state of the atmosphere is compared with new observations, then updated to reflect those observations in a new modeled state of the atmosphere. The process relies on an underlying climate model to combine disparate observations in a consistent manner, enabling production of gridded data sets for a broad range of variables, including ones that are sparsely measured or not directly observed. MERRA-2 is produced by the NASA Goddard Earth Observing System, Version 5 (GEOS-5) climate modeling system and comprises a global, temporally and spatially uniform synthesis of over 600 climate-related variables. Its native spatial resolution is 1/2° latitude × 5/8° longitude (i.e., 55.5 × 69.4 km at the equator) × 72 vertical levels extending through the stratosphere. MERRA-2 extends from 1979 to the present, and the temporal resolution of its variables is hourly.

Seven of our 14 variables reflect key attributes of the microclimate that relate to the biological or ecological functioning of the species (Alcaraz-Segura et al., 2017; Regos et al., 2022; Schnase et al., 1991). These variables helped us understand the influence of rapidly changing conditions near the Earth's surface. We refer to these as MERRA-2 or M2 variables. The remaining seven variables were modeled after the classic bioclimatic variables commonly used in ENM (O'Donnell & Ignizio, 2012; Vega et al., 2017). We derived these using MERRA-2 temperature and precipita-

TABLE 1. NASA's Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2; M2) and MERRA-2-derived bioclimatic variables (MERRAclim-2; MC) used in the study.

Variable	Description*
MERRA-2 (M2) Microclimatic variables:	
EVPINTR	Plant surface/absorption interception loss energy flux (W/m^2)
EVPSOIL	Bare soil evaporation energy flux (W/m^2)
PREVTOT	Total re-evaporation/sublimation of precipitation ((kg/m^2)/s)
QV2M	2-meter specific humidity (kg/kg)
SPEED	Surface wind speed (m/s)
SWGNT	Surface net downward shortwave flux (W/m^2)
TAUTOT	Optical thickness of all clouds
MERRAclim-2 (MC) Macroclimatic variables:	
MC_Bio03	Isothermality (%)
MC_Bio05	Maximum temperature of the warmest month (°C)
MC_Bio08	Mean temperature of the wettest quarter (°C)
MC_Bio13	Precipitation of the wettest month (mm)
MC_Bio14	Precipitation of the driest month (mm)
MC_Bio15	Precipitation seasonality (%)
MC_Bio18	Precipitation of the warmest quarter (mm)

TABLE 2. NASA's Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2; M2) variable definitions.*

Variable	Definition
2D Atmospheric variables:	
QV2M	2-meter specific humidity (kg/kg) – The amount of water vapor contained in a unit amount of air, generally expressed as kg of water per kg of air.
2D Surface fluxes:	
SPEED	Surface wind speed (m/s) – The speed of wind flow near the Earth's surface, expressed in meters per second.
PREVTOT	Total re-evaporation/sublimation of precipitation ([kg/m^2]/s) – The amount of precipitation that evaporates (water to water vapor transition) or sublimates (snow or ice to water vapor transition) while falling through the atmosphere and fails to arrive at the land surface, expressed in mm per second. [1 kg of water spread over a square meter (kg/m^2) = 1 mm]
2D Surface and top-of-atmosphere radiation fluxes:	
SWGNT	Surface net downward shortwave flux (W/m^2) – An estimate of the total amount of shortwave (0.3-4.0 µm) radiative energy that reaches the Earth's surface, expressed in Watts per square meter. SWGNT is an important source of energy and influence on land-atmosphere and vegetation interactions. SWGNT has many applications in the general and applied sciences.
TAUTOT	Optical thickness of all clouds – A measure of attenuation of the light passing through the atmosphere due to the scattering and absorption by cloud droplets. TAUTOT is a dimensionless, monotonically increasing function that approaches zero as cloud thickness approaches zero.
2D Land surface variables:	
EVPSOIL	Bare soil evaporation energy flux (W/m^2) – The rate of radiant energy transfer when water evaporates from a saturated land surface, expressed in Watts per square meter.
EVPINTR	Interception loss energy flux (W/m^2) – The portion of precipitation that is returned to the atmosphere through evaporation from plant surfaces or absorbed by plants and does not reach the ground, expressed in Watts per square meter.

* Summaries derived from definitions provided in: J. R. Holton, J. A. Curry, and J. A. Pyle, Eds., *Encyclopedia of atmospheric sciences*. Amsterdam; Boston: Academic Press, 2003. Details about the MERRA-2 collection and variable naming conventions can be found in in: Bosilovich, M.G., R. Lucchesi, and M. Suarez. 2016. "MERRA-2: File Specification." *GMAO Office Note* 9 (Version 1.1): 1–73.

tion variables as described in Schnase *et al*. (2024). We refer to these as MERRAclim-2 or MC variables. These macroclimatic variables helped us understand the influence of widespread, yearly and seasonal patterns of climatic change.

Building the seasonal climatic suitability time series–In constructing the biweekly time series models, we established variable time specificity with respect to our binned species observations at two levels. To capture seasonal macroclimatic influences, we aggregated the seven MC bioclimatic variables to the 5-year spans. To do this, we used the average yearly values for the MC variables for each 5-year span. We needed to use averaged yearly values for these variables because the scope of the MC variables has a yearly basis (e.g., maximum temperature of the warmest month [MC_Bio05], mean temperature of the wettest quarter [MC_Bio08], etc.). To capture higher temporal resolution microclimatic influences, we aggregated the seven M2 ecological functional attributes to the biweekly intervals in each 5-year span. We computed averaged biweekly values for these variables using 3-hour-resolution data from the MERRA-2 collection.

Each biweekly model, therefore, drew on a total of 14 environmental variables representing a combination of climatic influences at both coarse and fine temporal scales: seven MC variables, with average values for each 5-year span, and seven M2 variables, with average values for each biweekly interval.

We built the biweekly models using the averaged results from three, 5-method ensembles composed of MaxEnt (Elith et al., 2011; S. J. Phillips et al., 2017), Boosted Regression Tree (Elith et al., 2008), Random Forest (Breiman, 2001), Generalized Linear Model (Nelder & Wedderburn, 1972), and Multivariate Adaptive Regression Spline (Leathwick et al., 2006). We used the R sdm package with default settings to perform the ensemble runs (Naimi & Araújo, 2016). We chose these methods based on their demonstrated performance with presence-only data (Araújo & New, 2007; Elith et al., 2006; Salas et al., 2017). We used a simple, unweighted average of the ensemble models to determine the final biweekly model results.

Measuring variable contributions and model performance–To gain insights into the influence of specific environmental variables, we used permutation importance (PI) to determine the relative contributions of variables to the overall time series and to the individual models themselves (S. J. Phillips et al., 2017; Searcy & Shaffer, 2016; Smith & Santos, 2020). We assessed model performance using the Area Under the Receiver Operating Characteristic (ROC) Curve (AUC) (Fielding & Bell, 1997), the True Skill Statistic (TSS) (Allouche et al., 2006), and Percent Correctly Classified (PCC) (Warren & Seifert, 2011). In evaluating model performance, we regarded AUC measures < 0.7 as low, 0.7–0.9 as moderate, and > 0.9 as high; similarly, we regarded TSS and PCC measures < 0.5 as poor, 0.5–0.8 as useful, and > 0.8 as good (Arenas-Castro et al., 2018).

RESULTS

Northwesterly seasonal shift in suitable conditions–The time series models in our seasonal analysis revealed a consistent northwesterly shift in suitable conditions across the 10 biweekly intervals in each of the eight 5-year spans of the 40-year study (Fig. 1). These shifts resulted from decreasing suitability in the southeastern extent of Cassin's Sparrow's range followed by increasing suitability in northwestern regions over the course of the breeding season. The proportions of the study area affected by positive and negative trends during the breeding season were 49.7 ± 5.6% and 27.6 ± 9.7%, respectively, a ratio that changed little over the span of the study (Fig. 2). The weighted centroid for suitability likewise demonstrated shifts in a northwesterly direction across the summer months. These seasonal shifts were characterized by a decrease in centroid longitude from approximately -100° to -106°, reflecting a westward movement from the arid-humid divide in central Texas (Seager et al., 2008, 2018) toward the Chihuahuan Desert region of southern New Mexico. Seasonally, centroid longitude decreased by 4.7 ± 0.2° ($\beta = -0.7$; 95% CI: −0.9 to −0.5; $R^2 = 0.9$), based on model estimates averaged across the 40-year study period (Fig. 3). There were no statistically significant changes in the bearing or latitudinal components of the centroid shifts.

Macroclimatic and microclimatic variables drive shifting conditions–MC_Bio03 (isothermality), MC_Bio15 (precipitation seasonality), and MC_Bio05 (maximum temperature of the warmest month) were the most important overall contributory MC variables, all appearing in more than 50% of the model runs and having an average model PI of 20.6 ± 0.9%, 16.61 ± 0.9%, and 19.8 ± 1.0%, respectively (Fig. 4). EVPINTR (interception loss energy flux), SWGNT (surface net downward shortwave flux), SPEED (surface wind speed), and EVPSOIL (bare soil evaporation energy flux) were the most important overall contributory M2 variables with average model PIs of 16.9 ± 0.8%, 17.0 ± 0.9%, 29.2 ± 1.2%, and 18.7 ± 1.2%, respectively. A total of n = 1,200 model runs contributed to the climatic suitability time series (5 methods × 3 iterations × 10 biweekly intervals × eight 5-year spans), which attained an overall performance of AUC 0.83 ± 0.01, TSS 0.61 ± 0.02, and PCC 0.77 ± 0.01.

Decreasing seasonal shift distances over the past 40 years–Over the past four decades, the average distance traversed by the seasonal centroid for climatic suitability was 613.0 ± 76.6 km; however, this distance has decreased from approximately 922.1 km in the mid-1980s to 304.2 km in recent years ($\beta = -17.7$; CI: -30.3, -4.8; $R^2 = 0.6$; Figs. 2, 5). Northwesterly movement of the April centroid toward a relatively invariant August centroid largely accounts for the reduced traversal distances, suggesting that deteriorating conditions in the southeastern areas of Cassin's Sparrow's range might be influencing this long-term pattern (Fig. 2). These distances correspond to an average movement speed of 30.6 ± 3.8 km per week, which has likewise decreased from 46.1 km/wk to 15.2 km/wk over the same period ($\beta = -0.79$; 95% CI: −1.51 to −0.24; $R^2 = 0.6$), an overall rate of decline of approximately 0.79 km/wk/year.

Regional variation in the size of seasonal shifts–Our spatial analysis of the magnitude of seasonal shifts in climatic suitability shows increasing trends over approximately 62.1% of Cassin's Sparrow's range and decreasing trends over about 20.1% of the range over the past 40 years (Fig. 6). The amount of change over the course of the breeding season is increasing primarily in the central and western parts of the

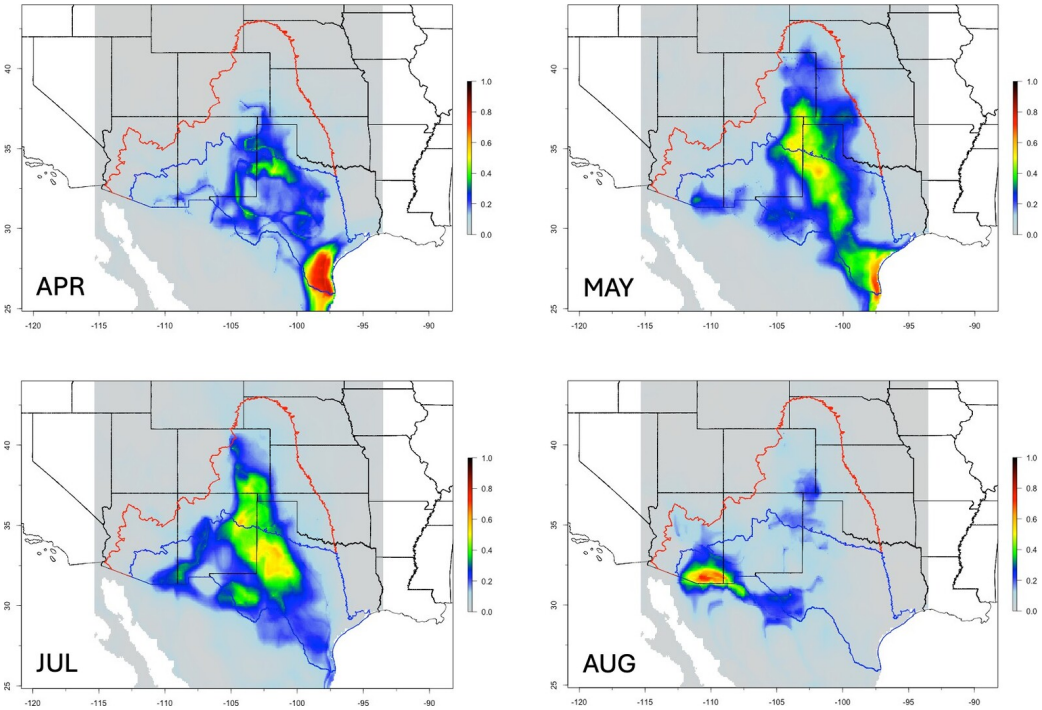

FIG. 1. Modeled seasonal patterns of climatic suitability for Cassin's Sparrow (*Peucaea cassinii*). Monthly maps show the east-to-west progression of suitable conditions during the breeding season (April–August) over the 5-year span from 2000 to 2004. Suitability values range from 0.0 to 1.0, with warmer colors indicating higher probabilities of suitable conditions. Red and blue lines denote the northern extent of Cassin's Sparrow's breeding and overwintering ranges, respectively.

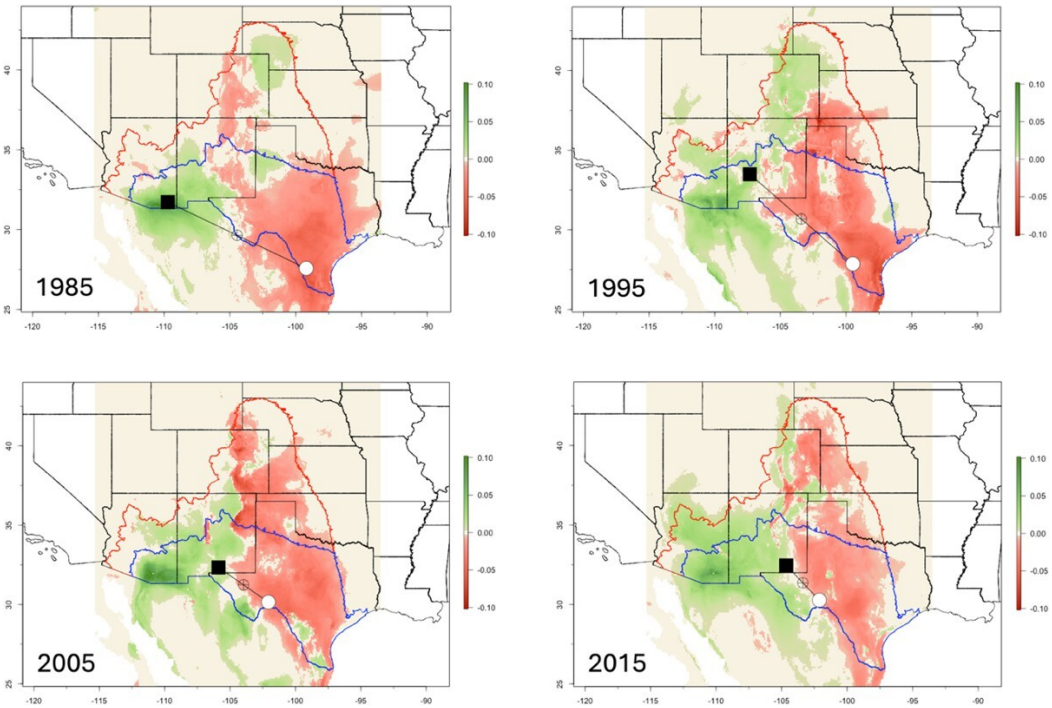

FIG. 2. Modeled seasonal climatic suitability shifts for Cassin's Sparrow (*Peucaea cassinii*) over the past four decades. Spatial trends are shown for selected 5-year intervals during the 40-year study period (1980–2020). Green areas indicate positive trends (increasing seasonal suitability), and red areas indicate negative trends (decreasing suitability). Color intensity reflects the rate of change in suitability probabilities over the course of the breeding season per 5-year interval. Vectors depict the movement of the seasonal suitability centroid from April (○) to August (■). Red and blue lines denote the northern extent of Cassin's Sparrow's breeding and overwintering ranges, respectively.

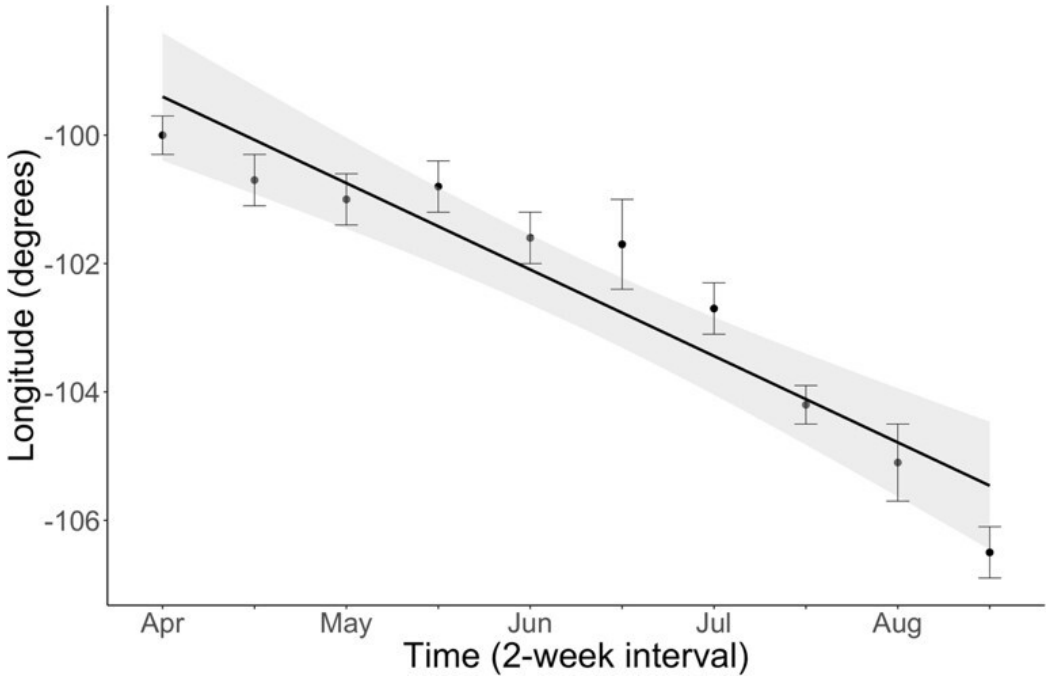

FIG. 3. Seasonal westward shift in the centroid of climatic suitability for Cassin's Sparrow (*Peucaea cassinii*) based on modeled estimates across the 40-year study period. Points represent mean estimated longitudes for each 2-week interval, with vertical bars indicating standard errors from the Bayesian regressions. The solid line shows the predicted trend in mean longitude over time, and the shaded area represents the 95% credible interval (2.5^{th}–97.5^{th} percentiles) of the predicted values. This seasonal shift extends from central Texas to southeastern New Mexico.

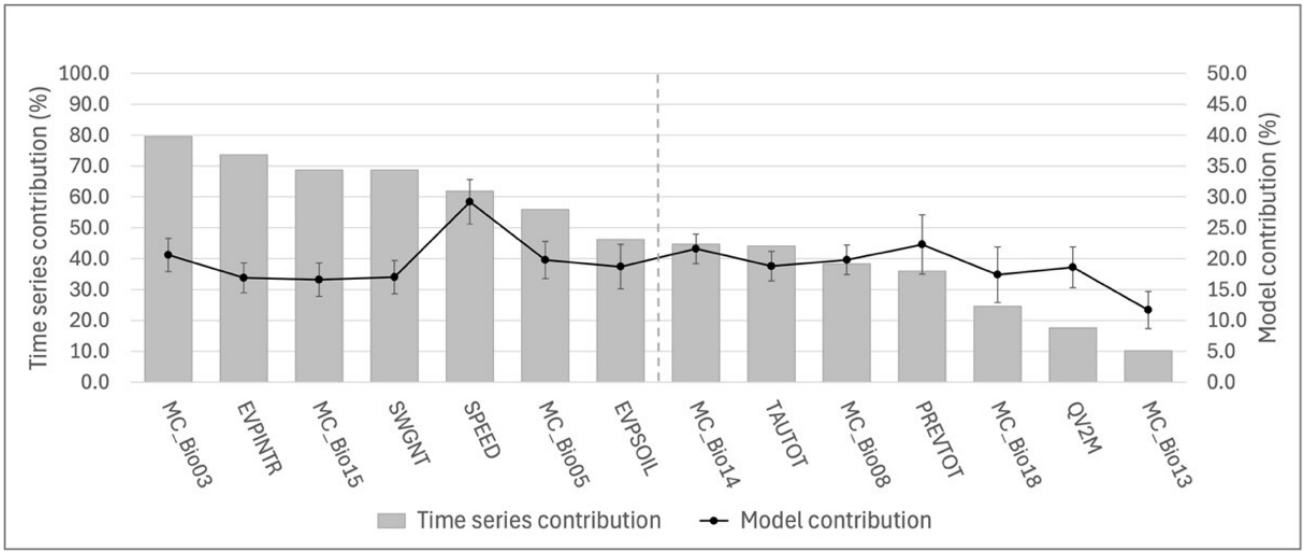

FIG. 4. Relative contribution of microclimatic and macroclimatic variables to the time series of modeled climatic suitability for Cassin's Sparrow (*Peucaea cassinii*). Rankings are based on the average permutation importance (PI) of each variable across individual models in the time series. Gray bars represent the percentage of the 1,200 models in which each variable contributed to the final model; black points and the connecting line show the average PI for each variable, with vertical bars indicating standard errors. Variables to the left of the vertical dashed line were the most influential overall, each appearing in more than 50% of the models.

Cassin's Sparrow's overall range; the decreases in this magnitude of change are occurring primarily in central Texas and along the southeastern coastal regions of that state. Scattered areas throughout the range show little or no change in the magnitude of seasonal shifts.

DISCUSSION

Cassin's Sparrows are notoriously difficult to detect in the field. They are rather plain, ground-dwelling birds that are generally quiet until the breeding season when males perform their distinctive flight song displays. This skylark-

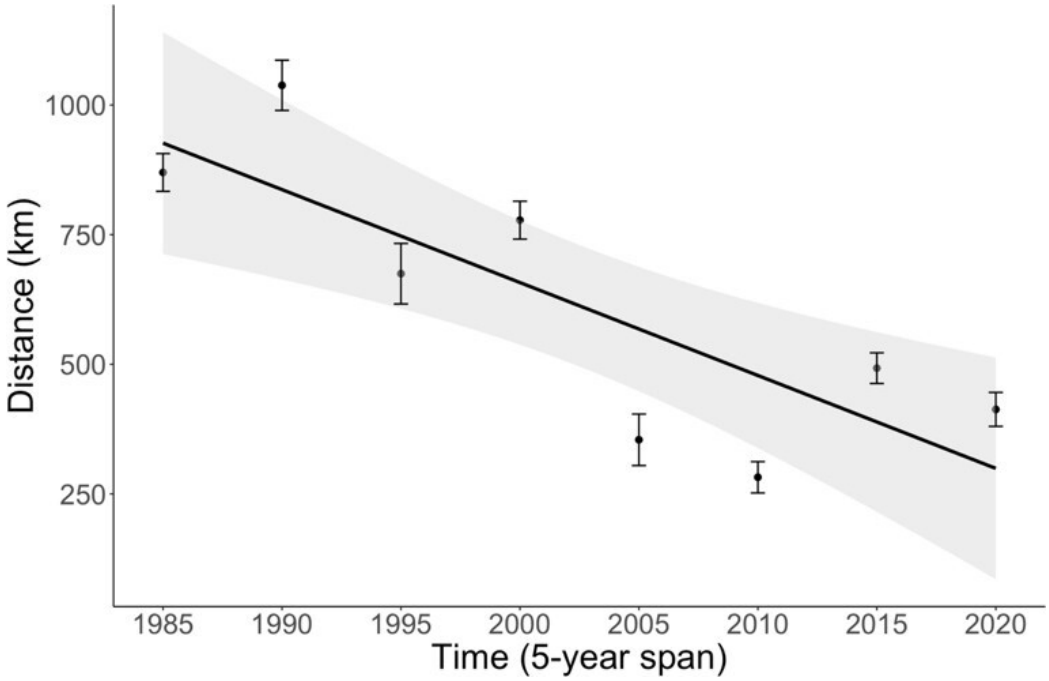

FIG. 5. Long-term decline in the distances associated with seasonal shifts in climatic suitability for Cassin's Sparrow (*Peucaea cassinii*). Points represent the mean distances (km) traversed by the weighted centroids of modeled climatic suitability over the breeding season, calculated for successive 5-year spans from 1985 to 2020. Vertical bars show standard errors derived from the Bayesian regression models. The solid line indicates the predicted trend, and the shaded area represents the 95% credible interval (2.5th–97.5th percentiles) of predicted values.

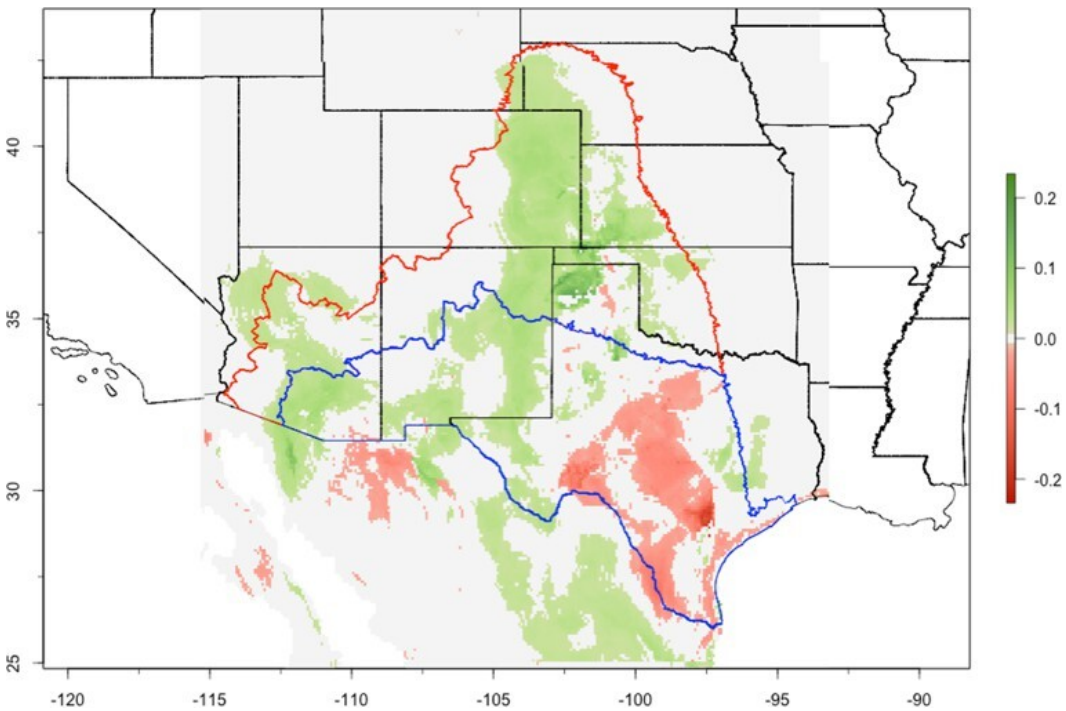

FIG. 6. Long-term spatial trends in the magnitude of seasonal shifts in climatic suitability for Cassin's Sparrow (*Peucaea cassinii*) over the past four decades. Unlike our use of color in Fig. 2, colors here show trends in the size of seasonal shifts in suitability for Cassin's Sparrow rather than changes in climatic suitability itself. Green areas indicate magnitude increases, red areas indicate magnitude decreases, and gray areas reflect little or no change. Red and blue lines show the northern limits of the Cassin's Sparrow's breeding and overwintering ranges, respectively. The relationship between the trends in magnitude of seasonal changes in suitability and trends in a region's overall climatic suitability remains unclear.

ing behavior is used to establish territories and court females (Dunning et al., 2020; Ruth, 2000; Schnase, 1984; F. C. Williams & LeSassier, 1968). Due to their elusive nature, much of what is known about the natural history of the species has come from indirect evidence. For example, song analysis has been used to identify individual males and map their territories (Schnase, 1984; Schnase & Maxwell, 1989), while energetic studies have provided estimates of various aspects of the reproductive ecology of the species (Schnase et al., 1991). More recently, eBird occurrence records have been used to quantify the breeding phenology for Cassin's Sparrow (Schnase et al., 2025). Here, we continue in this vein by describing what we believe are the most notable ecological insights gained by the current work.

East-to-west seasonal progression of suitable conditions–Our results reveal a spatially and temporally varying association between Cassin's Sparrow occurrences and climatic conditions that is consistent with an east-to-west progression in suitability over the summer breeding season. From 1985 through approximately 2000, suitable breeding conditions typically emerged in early spring within the coastal grass-brushlands of the Texas Rio Grande Valley (Fig. 2). As the season advanced, suitability in these eastern areas generally declined while improving farther west. By late summer, the most favorable conditions had shifted to the Chihuahuan and Sonoran Desert regions of New Mexico and Arizona. Mid-season to late-season improvements were also evident in the shortgrass and mixed-grass prairie regions of western Nebraska, eastern Colorado, northeastern New Mexico, and parts of the Texas Panhandle. Beginning around 2005, however, seasonal improvements in the northernmost prairie regions of the species' range gave way to seasonal declines in suitability. Collectively, these changes have produced a persistent east-to-west axis to the seasonal progression of climatic suitability for Cassin's Sparrow that remains evident today (Figs. 2, 3).

This pattern aligns with historical accounts and recent phenological studies (Dunning et al., 2020; Hubbard, 1977; Ohmart, 1969; Ruth, 2000; Schnase et al., 2025; F. C. Williams & LeSassier, 1968). It also supports the hypothesis that monsoon-driven precipitation influences the seasonal movements in this species. The North American Monsoon, which typically occurs during the summer months from late June through mid-September, strongly shapes summer rainfall patterns across the southwestern U.S. and northwestern Mexico (Adams & Comrie, 1997). Our results are consistent with Cassin's Sparrows responding to these precipitation events by shifting breeding activity westward into monsoon-affected regions later in the breeding season.

Wind speed, temperature and precipitation variability key environmental drivers–Previous studies have shown that increasing wind speeds, drying land surface conditions, and rainfall exert strong, year-round influences on environmental suitability for Cassin's Sparrow (Schnase et al., 2024). Variable contributions in the current intra-annual analysis confirm the importance of these drivers during the breeding season and highlight additional influences, particularly from variability in daily temperatures and monthly precipitation (Fig. 4). Among the macroclimatic drivers, isothermality (MC_Bio03), which measures the relative variation in daily versus annual temperatures, emerged as the top contributor to climatic suitability. Precipitation seasonality (MC_Bio15), which reflects the degree of variation in monthly precipitation across the year, was the third most influential variable. These results align with growing evidence that climatic variability per se is a key driver of avian phenological responses, likely mediated by species-specific ecological or physiological sensitivities (Albright et al., 2010; Andreasson et al., 2023; Shutt et al., 2022; Taff & Shipley, 2023), reductions in arthropod biomass, or other resource limitations (Dunn et al., 2023; Guralnick et al., 2023; Newell et al., 2023).

Collectively, the leading microclimatic drivers identified in this study highlight the importance of drying conditions during the summer breeding season in shaping climatic suitability for Cassin's Sparrow. These variables included interception loss energy flux (EVPINTR), which reflects water loss from plant surfaces; incoming shortwave radiation (SWGNT), representing solar energy input; surface wind speed (SPEED); and bare soil evaporation (EVPSOIL). Notably, wind plays a potentiating role across this suite of variables. By mixing sun-heated air near the ground, wind enhances the transfer of water from soil and vegetation surfaces into the atmosphere until drying conditions dominate (Lian et al., 2022; Tran et al., 2016). Future ENM efforts for Cassin's Sparrow would likely benefit from greater attention to these drying processes and the temporal variability of associated climatic variables. In light of the importance of aerial flight displays to Cassin's Sparrow's reproductive ecology, further investigation into the effects of surface wind speed on this behavior might also yield valuable insights.

Itinerant breeding possible for Cassin's Sparrow–Cassin's Sparrow appears to possess a capacity for facultative multiple breeding, a reproductive strategy in which individuals can produce more than one brood within a single breeding season (Hoffmann et al., 2015; Husby et al., 2009; Nomi et al., 2018). This has led to longstanding speculation that the species might be an itinerant breeder (Dunning et al., 2020; Hubbard, 1977; Ohmart, 1966, 1969; Ruth, 2000; F. C. Williams & LeSassier, 1968), a relatively rare phenomenon among North American birds (Baldassarre et al., 2019). If true, Cassin's Sparrows could either produce multiple clutches at a single site or relocate after initial breeding to more favorable conditions elsewhere for subsequent clutches, effectively establishing dual breeding territories within a season. In the absence of direct evidence, a central question has remained: is such itinerancy feasible?

Over the past 40 years, we observed seasonal centroid shifts in climatic suitability at rates ranging from 46.1 km/wk in 1985 to 15.2 km/wk in 2020, with a 40-year average of 30.6 km/wk. Given that Cassin's Sparrow's breeding cycle, from courtship to fledging, ranges from 5-6 weeks (Dunning et al., 2020; Ruth, 2000; Schnase et al., 1991), the optimum zone of climatic suitability could shift more than 275 km (46.1 km/wk × 6 wks) during that time span. For context, the Grasshopper Sparrow (*Ammodramus savannarum*), a species with similar physical, physiological, and life history traits, is known to make migratory movements exceeding 500 km/wk (Hill & Renfrew, 2019). Although our findings do not definitively confirm itinerant breeding in Cassin's Sparrow, the temporal and spatial scales of observed shifts in cli-

matic suitability suggest that such a strategy is feasible. Over an approximate 5-month long breeding season, individuals would have had ample time to produce multiple broods and to make opportunistic moves between clutches to track suitable conditions.

Regional adaptation preempting opportunistic movements–In our models, the weighted centroid for climatic suitability represents the theoretical center of a geographic area where optimal macroclimatic and microclimatic conditions exist at a given moment in time (Fig. 2). Seasonal shifts in these centroids presumably reflect the dynamic environmental conditions that Cassin's Sparrow tracks through its opportunistic movements. We observed a decline in the distances of these shifts over the past four decades (Figs. 2, 5), which suggests that suitable conditions for Cassin's Sparrow are either becoming more stable or that Cassin's Sparrow is increasingly able to adjust to ongoing environmental changes.

It is unlikely that climatic conditions are stabilizing, given extensive evidence to the contrary (Bateman, Pidgeon, Radeloff, VanDerWal, et al., 2016; Pacifici et al., 2015; Riddell et al., 2021; Wilsey et al., 2019) and work specifically showing a long-term shift in climatic suitability for Cassin's Sparrow (Schnase et al., 2024). Adjustments to changing environmental conditions are a more likely explanation. We know that birds can adapt to changing conditions through evolutionary shifts in physiological tolerance or by expressing flexible phenotypic responses that maintain functional physiological associations with their environments. Such responses include spatial shifts (e.g., in latitude or elevation) or temporal adjustments (e.g., in phenology) (Both & Visser, 2005; Charmantier & Gienapp, 2014; Neate-Clegg et al., 2024; Radchuk et al., 2019; Rollinson et al., 2021; Socolar et al., 2017; Tingley et al., 2009). Our decadal analysis reveals that the magnitude of seasonal shifts in climatic suitability has increased over the past 40 years in the shortgrass prairie regions of Cassin's Sparrow's central range (Fig. 6). In contrast, the magnitude of seasonal shifts has become less pronounced in central and south Texas. At present, the relationship between the magnitude of these changes and a region's overall climatic suitability for Cassin's Sparrow remains unclear. Recent studies, however, suggest that breeding phenology in this species varies regionally in ways that correspond with the spatial patterns of increasing and decreasing magnitudes of seasonal shifts documented in this study. Schnase *et al*. (2025) found that phenological advancement was most evident where our current analysis shows increasing seasonal changes, whereas delayed starts and advancing end phases appear to be contributing to substantial declines in breeding season duration in areas associated with decreasing seasonal changes. The apparent deterioration of conditions in the southeastern portion of Cassin's Sparrow's range, evidenced by the long-term northwesterly movement of the southern centroid for climatic suitability toward the relatively stable northern centroid in Fig. 2, may be contributing to the observed contraction of the breeding season in this southeastern region.

One possible driver for this apparent deterioration of conditions in the southeastern portion of Cassin's Sparrow's range is shifting vegetation types (Andersen & Steidl, 2022; Coffman et al., 2014; Ruth, 2000; Schooley et al., 2021; Schreiner-McGraw et al., 2020). The arid shrub-grasslands that define much of Cassin's Sparrow's North American range support avifauna whose spatial use of the landscape is highly sensitive to concomitant changes in vegetation structure (Iknayan & Beissinger, 2018; Olechnowski et al., 2009; Toenies et al., 2018). The ongoing shift from grass to shrubs across this broad regional vegetation gradient (Bestelmeyer et al., 2018; Wojcikiewicz et al., 2024) can significantly alter ecosystem processes, habitat suitability, breeding, and foraging behavior across multiple trophic groups (Coffman et al., 2014; Schooley et al., 2021; Schreiner-McGraw et al., 2020; Wagnon et al., 2020) and among grassland birds in particular (Andersen & Steidl, 2022; Anteau et al., 2023; Comer et al., 2018; Grand et al., 2019). This dynamic of shifting low-stature vegetation has been observed within other global biome boundaries, including the boreal forest (taiga)-tundra ecotone in the high northern latitudes (Montesano et al., 2020, 2024), and is likely to influence Cassin's Sparrow as well.

At this point, it is impossible to say whether the patterns we see are the result of local adaptation, phenotypic plasticity, climate-related or anthropogenic changes in vegetation structure or land cover, or other phenomenon. Nonetheless, these results support a recent hypothesis that adaptation to regional environmental differences might be playing an increasingly important role in Cassin's Sparrow's response to long-term climate change (Schnase et al., 2025). Additionally, the apparent association between the magnitude of seasonal changes in climatic suitability and reported shifts in breeding phenology raises the possibility that the magnitude of seasonal variability itself might be a driver of phenological change in Cassin's Sparrow. Given these complexities, our understanding of the conservation status of Cassin's Sparrow and potential use of the species as a representative of aridland ecosystem conditions in broader conservation efforts would benefit from further study.

Limitations and future directions–Although we took steps to address spatiotemporal biases in the preparation and analysis of observational data, these results still should be interpreted with some caution. As with most large-scale occurrence datasets, particularly those derived from citizen-science initiatives such as eBird, residual sampling effects associated with changes in observer participation over time may persist and influence the spatial patterns based on ENM results. In addition, seasonal and geographic differences in detectability, observer effort, and accessibility may differentially influence occurrence records, potentially confounding comparisons of modeled suitability across space or time (Johnston et al., 2021).

The climate-occurrence relationships identified by our models are consistent with the hypothesis that Cassin's Sparrow tracks seasonally shifting climatic conditions; however, different combinations of climatic conditions can produce similar estimates of environmental suitability, and climatic variables often function as proxies for unmeasured environmental variables or complex ecological processes (Dormann et al., 2012). As a result, modeled suitability alone cannot distinguish between similar and distinct environmental conditions across space or time, nor can it establish causal relationships in the absence of supporting

physiological, behavioral, or mechanistic evidence (Broennimann et al., 2012; Gómez et al., 2016; Illán et al., 2022). Future work integrating experimental or physiological data, finer-resolution habitat information, and independent measures of demographic performance would help clarify the mechanisms underlying the observed patterns. In particular, linking climate-associated suitability to survival, reproduction, resource availability, and vegetation condition would provide stronger evidence for climate tracking and improve understanding of this species' response to ongoing and future environmental change.

CONCLUSIONS

In this study, we used rENM, high temporal-resolution environmental data from NASA's MERRA-2 reanalysis, and 40 years of eBird observations to improve our understanding of Cassin's Sparrow's breeding ecology. We selected Cassin's Sparrow as a focal species due to its relevance as an aridland and grassland indicator species and its capacity for opportunistic breeding, with movements and reproductive activity influenced by favorable environmental conditions rather than a fixed schedule. Our analyses confirmed a previously hypothesized east-to-west seasonal progression in climatic suitability across the summer breeding season, driven largely by surface wind speed and variability in temperature and precipitation. These shifts averaged ~613.0 km over the breeding season and have diminished over the past 40 years. If Cassin's Sparrow is indeed an itinerant breeder, as has been long speculated, it is plausible that individuals can track these shifting conditions over the traversal distances we observed and produce multiple broods during the course of the breeding season. We also found geographic variation in the magnitude of seasonal changes in climatic suitability and evidence linking this variation to changes in breeding phenology reported in other studies. This suggests that the magnitude of seasonal change itself might drive phenological responses in the species and that adaptation to regional climatic variation might be more important now and into the future than the opportunistic movements observed in the past. These findings provide a more complete view of the seasonal weather patterns and long-term climatic conditions affecting Cassin's Sparrow's breeding ecology than heretofore reported. Beyond improving our understanding of this one species, however, the work demonstrates a promising way of studying the multiscale dynamics of climatic influences on grassland and aridland birds more broadly.

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